Chapter 1 — The Spring Ecosystem and Your First App

Spring is the most widely used framework for server-side Java applications (programs that run on a server and answer other programs): web APIs, back-office systems, batch jobs (programs that process a large pile of data in one run, often at night) and message consumers (programs that react to messages arriving on a queue).

> **Key term — API / REST API:** An **API** (Application Programming Interface) is the set of operations one program offers to others. A **web API** offers them over HTTP. A **REST API** is a web API in a common style: each thing (an order, a user) has a URL, and you act on it with standard HTTP methods such as `GET` (read) and `POST` (create), usually exchanging JSON. HTTP basics: System Design Fundamentals.

This chapter explains what Spring and Spring Boot are, their history and current versions, then builds, runs, packages and tests a first REST API with **Spring Boot 4.1.1** on **Java 25**. You only need basic Java; Java features (records, annotations, lambdas, Maven) are linked to the Java Handbook rather than re-taught.

All runnable code is in `code/hello-boot/`: a Maven project whose tests were run with `mvn -q -B -ntp verify` on JDK 25. Outputs below are real (long log lines trimmed with `...`).

1.1 What Spring is

What it is

A **library** is code you call (e.g. a JSON parser). A **framework** is code that calls *you*: it owns the program's structure and plugs your classes in at the right moments. This reversal is **Inversion of Control (IoC)** — "don't call us, we'll call you".

Spring Framework is an open-source Java framework whose core is an **IoC container**: an object that creates your application's objects, connects them, configures them and manages their whole life. Objects managed by the container are **beans**. Around that core Spring offers modules for web (Spring MVC — the classic web framework, one thread per request; WebFlux — a non-blocking alternative), data access (**JDBC** — Java Database Connectivity, Java's standard low-level SQL API; and **transactions** — groups of database changes that succeed or fail as one unit), messaging, testing, scheduling and more.

The problem it solves

Without a container, every class builds its own dependencies with `new`:

```
// Hard-wired: OrderService decides WHICH repository and WHICH mailer it uses.
class OrderService {
    private final OrderRepository repo = new JdbcOrderRepository(new HikariDataSource(...));
    private final Mailer mailer = new SmtpMailer("smtp.example.com", 25);
}
```

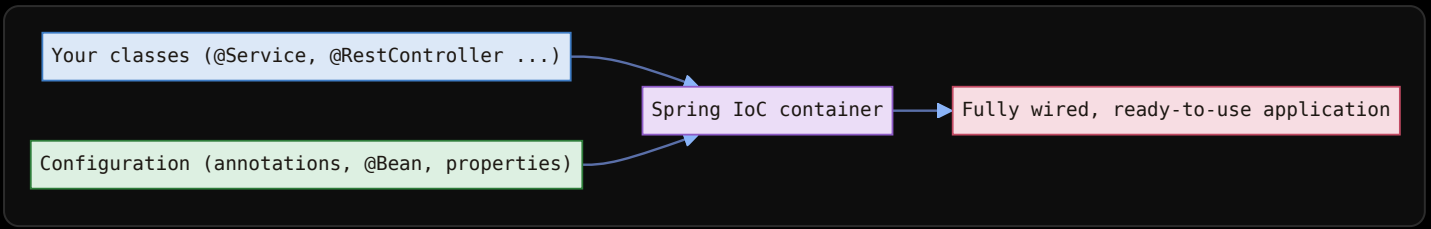
Problem	Why it hurts
Tight coupling	Swapping `JdbcOrderRepository` for another implementation means editing `OrderService`.
Hard to test	A unit test of `OrderService` now needs a real database and a real mail server.
Scattered configuration	Host names, pool sizes and credentials are buried in constructors all over the code.

Spring's answer is **Dependency Injection (DI)**: a class *declares* what it needs (usually as constructor parameters) and the container *supplies* it. The class depends on an interface; the container picks the implementation. Chapter 2 covers DI in depth; the design principle behind it is the Dependency Inversion Principle — see LLD §4.10 and the Hollywood Principle in LLD Chapter 5.

Beyond DI, Spring gives battle-tested answers for the plumbing every server needs: HTTP handling, JSON conversion (**JSON** is a plain-text data format like `{"name":"Ada"}`), transactions, security, scheduling, caching and consistent exception types.

How it works (bird's-eye view)

You write ordinary Java classes (POJOs — "Plain Old Java Objects") and mark those Spring should manage, e.g. with `@Service` (an **annotation**: a `@`-label frameworks read at runtime). At startup the container reads this **configuration metadata** (annotations, `@Bean` methods, or XML in old projects), creates each bean injecting what it needs, and may wrap beans in **proxies** (stand-ins that intercept calls) to add transactions or security (Chapter 4). It then calls your code when requests, messages or timers arrive, and at shutdown runs clean-up callbacks.



Real-world examples

Banks, retailers, airlines, streaming and public-sector back ends; Netflix has publicly described moving its Java services to Spring Boot. It is the most common Java stack in back-end interviews.

1.2 What Spring Boot adds

What it is

Spring Boot is built *on top of* Spring Framework to make Spring applications quick to create and production-ready. It does not replace Spring — every Boot application *is* a Spring application.

Feature	What it means	Where in this book
Starters	One dependency per capability (`spring-boot-starter-webmvc`) bringing a tested set of libraries.	§1.7
Dependency management	Boot picks compatible versions of hundreds of libraries, so your `pom.xml` has no `<version>` tags.	§1.6
Auto-configuration	Boot inspects your classpath (the jars your app includes) and settings and creates sensible beans (web server, JSON mapper, a `DataSource` — an object that hands out database connections). Your own beans always win.	Chapter 3
Embedded server	Tomcat (or Jetty) runs *inside* your app; you run `java -jar app.jar`.	§1.9
Externalized configuration	Settings from `application.properties`/`.yaml`, environment variables (e.g. `SERVER_PORT=9090`), command-line arguments, with a clear precedence order.	Chapter 3
Production features	Health checks, metrics, tracing (following one request across services), structured logging, graceful shutdown (Actuator).	Chapter 15
Testing support	Slices (tests that start only one layer) like `@WebMvcTest`, `@SpringBootTest`, test starters.	Chapter 12

Why it matters

Before Boot (2014), a Spring web app needed a WAR file (Web Application Archive — a zip you copy into a separately installed server), an external Tomcat, configuration for the `DispatcherServlet` (Spring MVC's central request dispatcher, Chapter 5), a view resolver, a JSON converter, a data source, and a careful

choice of mutually compatible library versions. Boot reduced that to a generated project and a `main` method, keeping every default overridable.

Spring vs Spring Boot

Question	Spring Framework	Spring Boot
What is it?	The core: IoC container, MVC, transactions...	An opinionated layer (sensible defaults) that configures Spring for you
Use one without the other?	Yes — plain Spring works without Boot	No — Boot always uses Spring Framework
Config / server / versions	You declare infrastructure beans, deploy a WAR, pick each library version	Auto-configured, embedded server, versions from Boot's BOM
Typical use today	Libraries, special setups	Almost every new application

Common mistakes

- Thinking Boot is a different framework with different annotations. `@Service`, `@Autowired`, `@Transactional`, `@RestController` are Spring *Framework* annotations; Boot mostly adds `@SpringBootApplication`, `@ConfigurationProperties` and test annotations.
- Believing auto-configuration is uninspectable magic. It is ordinary `@Configuration` classes guarded by conditions, and Boot can print a report of every decision (Chapter 3).

In the interview

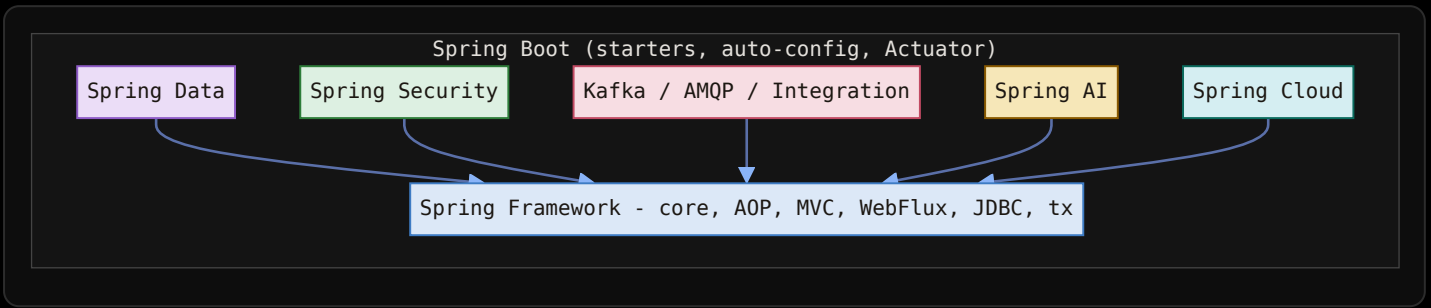
Q: What is the difference between Spring and Spring Boot? *30-second answer:* Spring Framework provides the IoC container and modules like MVC and transactions. Boot sits on top: starters for dependency sets, a managed version BOM, auto-configuration that creates beans based on classpath and properties, an embedded server so the app runs as a plain jar, externalized configuration and production features like Actuator. Boot removes boilerplate; it does not replace Spring.

- *Follow-up: Can you override what Boot configures?* Yes. Auto-configured beans are usually guarded by `@ConditionalOnMissingBean`, so defining your own bean of that type makes Boot back off; many settings are just properties.
- *Follow-up: Does Boot make the app slower?* Startup does extra condition evaluation, but the runtime path is the same Spring code. Boot also offers AOT (ahead-of-time: work done at build time instead of startup) processing and native images (the app compiled to a stand-alone executable needing no JVM) for faster startup (Chapter 15).

1.3 The Spring ecosystem

"Spring" is also an umbrella name for a family of projects maintained by the Spring team (part of Broadcom since its acquisition of VMware). All are version-managed by Spring Boot, so you rarely pick their versions yourself.

Project	What it gives you	Chapter
Spring Framework	IoC container, AOP (Aspect-Oriented Programming — cross-cutting behaviour such as logging around methods), MVC, WebFlux, JDBC, transactions, testing	2, 4, 5, 7, 13
Spring Boot	Starters, auto-configuration, embedded servers, Actuator	1, 3, 12, 15
Spring Data	Repository interfaces (you declare data-access methods; Spring writes the code) for JPA, JDBC, MongoDB, Redis, R2DBC...	8, 9
Spring Security	Authentication (who are you?), authorization (what may you do?), OAuth2 (delegated login/access), JWT (a signed token carrying user claims)	10, 11
Spring Cloud	Config server, service discovery, gateway (single entry point in front of many services), load balancing	16
Spring for Apache Kafka / Spring AMQP	Messaging with Kafka and RabbitMQ (message brokers)	14
Spring Batch	Large, restartable batch jobs	— (mentioned in 14)
Spring Integration	Enterprise integration patterns (channels, adapters)	—
Spring Session	Store HTTP sessions (per-user server-side state) in Redis or a database	10
Spring for GraphQL	GraphQL (query language where the client picks the fields) servers	—
Spring Modulith	Structure a monolith (one app deployed as a single unit) into verified modules	16
Spring AI	LLM chat clients, embeddings (text as number vectors), RAG (retrieval-augmented generation), tool calling	17



Real-world stack. A typical online-shop order service: Spring Boot + Spring MVC (REST API) + Spring Data JPA (PostgreSQL) + Spring Security (JWT) + Spring for Apache Kafka ("order placed" events) + Actuator/Micrometer (Spring's metrics library). This book builds each layer.

1.4 History and versions

A short timeline

Year	Release	What changed
2002	Rod Johnson's book *Expert One-on-One J2EE Design and Development*	A lighter alternative to EJB (Enterprise JavaBeans, J2EE's heavyweight component model); its sample code became Spring
2004	Spring Framework 1.0	IoC container, AOP, transaction abstraction, XML configuration
2007	Spring 2.5	Annotation-driven config: `@Autowired`, `@Component`, `@Controller`
2009	Spring 3.0	Java 5 baseline, `@Configuration` / `@Bean`, SpEL (Spring Expression Language, `#{...}`), REST support in MVC
2013	Spring 4.0	Java 8 support, `@Conditional` (foundation of auto-configuration)
2014	Spring Boot 1.0	Starters, auto-configuration, embedded servers
2017–18	Spring 5.0 / Boot 2.0	Java 8 baseline, reactive stack (WebFlux)
2022	Spring 6.0 / Boot 3.0	**Java 17 baseline**, `javax.*` → `jakarta.*`, AOT and GraalVM native images
Nov 2025	**Spring Framework 7.0** (13 Nov) / **Spring Boot 4.0** (20 Nov)	Jakarta EE 11, JSpecify null-safety (annotations marking what may be `null`), modular Boot jars, Jackson 3, API versioning, HTTP service clients, JUnit 6
Jun 2026	Spring Boot 4.1.0 (10 Jun)	gRPC (fast binary remote-call protocol) support, Jackson config updates, observability updates
Aug 2026	Spring Boot 4.1.1 (20 Aug)	Bug fixes and dependency upgrades — **the version used in this book**

The current generation — what Boot 4.1.1 gives you

From Boot 4.1.1's dependency BOM (`spring-boot-dependencies-4.1.1.pom`):

Component	Version in Boot 4.1.1	Notes
Java	**17 minimum**, 25 recommended	This book uses Java 25 (LTS) — see Java Ch. 17
Spring Framework	7.0.9	
Jakarta EE	11 (Servlet 6.1; JPA 3.2 — object/database mapping; Bean Validation 3.1 — rules like `@NotNull`)	
Tomcat / Jetty	11.0.24 / 12.1.12	Undertow no longer supported (lacks Servlet 6.1)
Jackson	3.1.5 (`tools.jackson` packages)	Jackson 2 still available via a separate, deprecated module
Hibernate ORM / Validator	7.4.5 / 9.1.3	ORM = object-relational mapper
Spring Security	7.1.1	
Spring Data	2026.0.1 release train (modules released together)	
JUnit Jupiter	6.0.3	JUnit 4 support deprecated in Framework 7
Mockito / AssertJ	5.23.0 / 3.27.7	Test fakes / readable assertions
Kotlin	2.3.21 (2.2 minimum)	
GraalVM (native images)	25+	

`javax` → `jakarta`

Java EE was handed to the Eclipse Foundation in 2017 and renamed **Jakarta EE**; for trademark reasons its packages moved from `javax.*` to `jakarta.*` (Jakarta EE 9, 2020). Since Spring 6 / Boot 3 Spring uses only `jakarta` names (`jakarta.servlet.http.HttpServletRequest`, `jakarta.persistence.Entity`, `jakarta.annotation.PostConstruct`, `jakarta.validation.constraints.NotNull`). Framework 7 removed the last support for `javax.annotation` and `javax.inject`. A tutorial importing `javax.persistence` is for Boot 2 or older.

How versions relate

- **Boot pins everything.** You choose the Boot version; its BOM chooses compatible Spring Framework, Data, Security, Tomcat, Jackson, Hibernate, JUnit, etc.
- **Cadence.** A new minor (4.0, 4.1 — features, no breaking changes) roughly every six months; patch releases (4.1.1, 4.1.2 — bug fixes only) in between.
- **Support.** Each minor line gets roughly 13 months of free open-source support, then commercial support. Check spring.io/projects/spring-boot "Support" before planning an upgrade.
- **Version names.** `-M1`, `-M2` milestones, `-RC1` release candidates, plain `4.1.1` is GA (generally available). `-SNAPSHOT` = nightly builds — never in production.

Common mistakes

- Adding `<version>` to a Spring dependency in a Boot project — overrides the tested version. Let the parent manage it.
- Mixing `javax.*` and `jakarta.*` imports after copying old code.
- Following a Boot 2/3 tutorial for Boot 4 without checking renamed starters (§1.7) and Jackson 3 package names.

1.5 Creating a project with start.spring.io

What it is

Spring Initializr (**start.spring.io**) is a web service that generates a ready-to-build Boot project as a zip. IntelliJ IDEA, VS Code (Spring Boot Extension Pack) and Eclipse (Spring Tools) call the same service from their "New project" wizards.

How it works

Choose: **Project** — *Gradle - Groovy* (default), *Gradle - Kotlin* or *Maven* (build tools; this book uses Maven); **Language** — Java (Kotlin, Groovy offered); **Spring Boot** — latest GA without suffix (**4.1.1**; avoid `SNAPSHOT` / `M` / `RC`); **Metadata** — *Group* `com.example` (reversed domain), *Artifact* `hello-boot` (also the jar name), *Package* `com.example.hello` (main class goes here), *Packaging* `Jar` (`War` only for an external server), *Java* 17/21/25/27 (this book: 25); **Dependencies** — *Spring Web* (id `web`) and *Spring Boot DevTools* (id `devtools`). Then Generate, unzip, open. From a terminal:

```
curl https://start.spring.io/starter.zip \
  -d type=maven-project -d bootVersion=4.1.1 -d javaVersion=25 \
  -d groupId=com.example -d artifactId=hello-boot -d packageName=com.example.hello \
  -d dependencies=web,devtools -o hello-boot.zip
```

For Boot 4.1.1, *Spring Web* produces `spring-boot-starter-webmvc` and the test starter `spring-boot-starter-webmvc-test` — exactly the `pom.xml` in §1.6 (checked against the generated file).

Common mistakes

- Choosing a Java version newer than your installed JDK (`release version 25 not supported`). Check `java -version`, see Java §1.4.
- Adding every dependency "just in case". Each starter can switch on auto-configuration — e.g. *Spring Data JPA* without a database fails with "Failed to configure a DataSource".
- Putting classes in a package *outside* the main class's package (§1.8).

1.6 Project layout and the build file

The layout

```
hello-boot/
├── pom.xml                          ← build file (Maven)
├── mvnw, mvnw.cmd, .mvn/            ← Maven Wrapper: builds without installing Maven
└── src/
    ├── main/
    │   ├── java/com/example/hello/
    │   │   ├── HelloApplication.java   ← the main class (§1.8)
    │   │   ├── HelloController.java    ← REST endpoint (§1.11)
    │   │   ├── GreetingService.java
    │   │   └── Greeting.java
    │   └── resources/
    │       ├── application.properties  ← configuration (Chapter 3)
    │       ├── static/                 ← optional: files served as-is (HTML, CSS, JS)
    │       └── templates/              ← optional: server-side templates (Thymeleaf)
    └── test/java/com/example/hello/
        ├── HelloControllerTest.java
        └── HelloApplicationTests.java
```

Standard Maven layout — `src/main/java`, `src/main/resources`, `src/test/java` (Java §18.1–18.2). Everything under `src/main/resources` ends up on the **classpath** (where the JVM loads classes and files from), which is how Boot finds `application.properties`.

The `pom.xml`

```xml
<!-- code/hello-boot/pom.xml -->
<parent>                                              <!-- 1 -->
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-parent</artifactId>
  <version>4.1.1</version>
  <relativePath/>
</parent>

<groupId>com.example</groupId>
<artifactId>hello-boot</artifactId>
<version>0.0.1-SNAPSHOT</version>

<properties>
  <java.version>25</java.version>                     <!-- 2 -->
</properties>

<dependencies>
  <dependency>                                        <!-- 3 -->
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-webmvc</artifactId>
  </dependency>
  <dependency>                                        <!-- 4 -->
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-devtools</artifactId>
    <scope>runtime</scope>
    <optional>true</optional>
  </dependency>
  <dependency>                                        <!-- 5 -->
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-webmvc-test</artifactId>
    <scope>test</scope>
  </dependency>
</dependencies>

<build>
  <plugins>
    <plugin>                                          <!-- 6 -->
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
```

1. **`spring-boot-starter-parent`** — a parent POM (Project Object Model, Maven's `pom.xml`; children inherit its settings) that (a) imports the dependency BOM (**Bill of Materials**: tested versions for hundreds of libraries), (b) sets UTF-8, (c) compiles with `-parameters` (so Spring can read constructor parameter names), (d) pre-configures plugins (surefire for tests, resources filtering, the Boot plugin).
2. **`java.version`** — becomes `maven.compiler.release`, so `javac --release 25`.
3. **Web starter** — no `<version>`: the parent manages it.
4. **DevTools** — `runtime` + `optional` keeps it out of projects depending on yours; the Boot plugin also leaves it out of the packaged jar by default.
5. **Test starter** — test classpath only.
6. **`spring-boot-maven-plugin`** — adds goals such as `spring-boot:run`, and during `package` **repackages** the jar into an executable "fat jar" containing all dependency jars (§1.9).

If you cannot use the parent (company parent POM), import the BOM and configure `-parameters` and the plugin's `repackage` goal yourself:

```
<dependencyManagement>
  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-dependencies</artifactId>
      <version>4.1.1</version>
      <type>pom</type>
      <scope>import</scope>
    </dependency>
  </dependencies>
</dependencyManagement>
```

Gradle equivalent (what Initializr generates):

```
plugins {
    id 'java'
    id 'org.springframework.boot' version '4.1.1'
    id 'io.spring.dependency-management' version '1.1.7'
}
java { toolchain { languageVersion = JavaLanguageVersion.of(25) } }
dependencies {
    implementation 'org.springframework.boot:spring-boot-starter-webmvc'
    developmentOnly 'org.springframework.boot:spring-boot-devtools'
    testImplementation 'org.springframework.boot:spring-boot-starter-webmvc-test'
    testRuntimeOnly 'org.junit.platform:junit-platform-launcher'
}
tasks.named('test') { useJUnitPlatform() }
```

Commands: `./gradlew bootRun`, `./gradlew bootJar`, `./gradlew test`.

1.7 Starters

What it is

A **starter** is a dependency with (almost) no code; it pulls in a curated, version-aligned set of libraries for one capability. `spring-boot-starter-webmvc` gives you Spring MVC, embedded Tomcat, Jackson, logging, and the Boot auto-configuration that wires them.

How it works

Real runtime dependency tree of `hello-boot` (`mvn dependency:tree -Dscope=runtime`):

```
com.example:hello-boot:jar:0.0.1-SNAPSHOT
+- org.springframework.boot:spring-boot-starter-webmvc:jar:4.1.1:compile
|  +- org.springframework.boot:spring-boot-starter:jar:4.1.1:compile
|  |  +- org.springframework.boot:spring-boot-starter-logging:jar:4.1.1:compile   (Logback 1.5.38, SLF4J bridges)
|  |  ...
|  +- org.springframework.boot:spring-boot-starter-jackson:jar:4.1.1:compile
|  |  \- org.springframework.boot:spring-boot-jackson:jar:4.1.1:compile
|  |     \- tools.jackson.core:jackson-databind:jar:3.1.5:compile
|  +- org.springframework.boot:spring-boot-starter-tomcat:jar:4.1.1:compile
|  |  +- org.springframework.boot:spring-boot-starter-tomcat-runtime:jar:4.1.1:compile
|  |  |  +- org.apache.tomcat.embed:tomcat-embed-core:jar:11.0.24:compile
|  |  |  ...
|  |  \- org.springframework.boot:spring-boot-tomcat:jar:4.1.1:compile
|  +- org.springframework.boot:spring-boot-http-converter:jar:4.1.1:compile
|  |  \- org.springframework:spring-web:jar:7.0.9:compile
|  \- org.springframework.boot:spring-boot-webmvc:jar:4.1.1:compile
|     +- org.springframework.boot:spring-boot-servlet:jar:4.1.1:compile
|     \- org.springframework:spring-webmvc:jar:7.0.9:compile
\- org.springframework.boot:spring-boot-devtools:jar:4.1.1:runtime (optional)
   +- org.springframework.boot:spring-boot:jar:4.1.1:compile   (spring-core / spring-context 7.0.9, jspecify
1.0.1)
   \- org.springframework.boot:spring-boot-autoconfigure:jar:4.1.1:compile
```

The Boot 4 pattern: each technology has a **Boot module** holding its auto-configuration (`spring-boot-webmvc`, `spring-boot-tomcat`, `spring-boot-jackson`) and a **starter** combining that module with the library.

Boot 4 modular starters

Boot 3 shipped all auto-configuration in one `spring-boot-autoconfigure` jar. **Boot 4 split it into per-technology modules** and renamed some starters. In 4.1.1 the core `spring-boot-autoconfigure` jar lists only 12 auto-configurations; MVC's live in `spring-boot-webmvc`, Tomcat's in `spring-boot-tomcat`, etc. (Chapter 3).

Capability	Boot 4 starter	Boot 3 name (if different)
Servlet web / REST (Spring MVC + Tomcat)	`spring-boot-starter-webmvc`	`spring-boot-starter-web`
Reactive web (WebFlux + Netty)	`spring-boot-starter-webflux`	
Blocking HTTP client (`RestClient`)	`spring-boot-starter-restclient`	part of `-web`
Reactive HTTP client (`WebClient`)	`spring-boot-starter-webclient`	part of `-webflux`
JSON (Jackson 3)	`spring-boot-starter-jackson`	`spring-boot-starter-json` (still available in 4.x)
Bean Validation	`spring-boot-starter-validation`	
JDBC / JPA	`spring-boot-starter-jdbc` / `spring-boot-starter-data-jpa`	
Security	`spring-boot-starter-security`	
OAuth2 resource server	`spring-boot-starter-security-oauth2-resource-server`	`spring-boot-starter-oauth2-resource-server`
AspectJ-based AOP	`spring-boot-starter-aspectj`	`spring-boot-starter-aop`
Kafka	`spring-boot-starter-kafka`	(previously just `spring-kafka`)
Actuator	`spring-boot-starter-actuator`	
Core only (no web)	`spring-boot-starter`	
Testing, generic	`spring-boot-starter-test`	
Testing, per technology	`spring-boot-starter-webmvc-test`, `-data-jpa-test`, `-security-test`, ...	— (new in Boot 4)

Rules of thumb for Boot 4:

- **Every technology starter has a `-test` companion** bringing its test slice (`@WebMvcTest` lives in `spring-boot-webmvc-test`). Each test starter includes `spring-boot-starter-test`.
- Some old names remain in 4.1 as **deprecated** aliases (`spring-boot-starter-web`, `spring-boot-starter-oauth2-resource-server`); others are gone (`spring-boot-starter-aop` → `spring-boot-starter-aspectj`). Prefer the new names.
- For a quick Boot 3 upgrade, `spring-boot-starter-classic` and `spring-boot-starter-test-classic` restore the old "everything on the classpath" behaviour while you fix imports.
- Third-party starters are named `<name>-spring-boot-starter` (e.g. `mybatis-spring-boot-starter`); only official ones start with `spring-boot-starter-` (Chapter 3 builds one).

Common mistakes

- Adding both `-webmvc` and `-webflux` starters and being surprised it's a *servlet* app: with both present Boot chooses Spring MVC.

- Using `spring-boot-starter-test` alone and finding `@WebMvcTest` missing in Boot 4 — add `spring-boot-starter-webmvc-test`.
- Excluding a transitive library (one you get indirectly) by hand to "slim down" and breaking auto-configuration. Use the intended starter (e.g. exclude `spring-boot-starter-tomcat`, add `spring-boot-starter-jetty`).

1.8 The main class and `@SpringBootApplication`

```java
// code/hello-boot/src/main/java/com/example/hello/HelloApplication.java
package com.example.hello;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication            // = @SpringBootConfiguration + @EnableAutoConfiguration + @ComponentScan
public class HelloApplication {

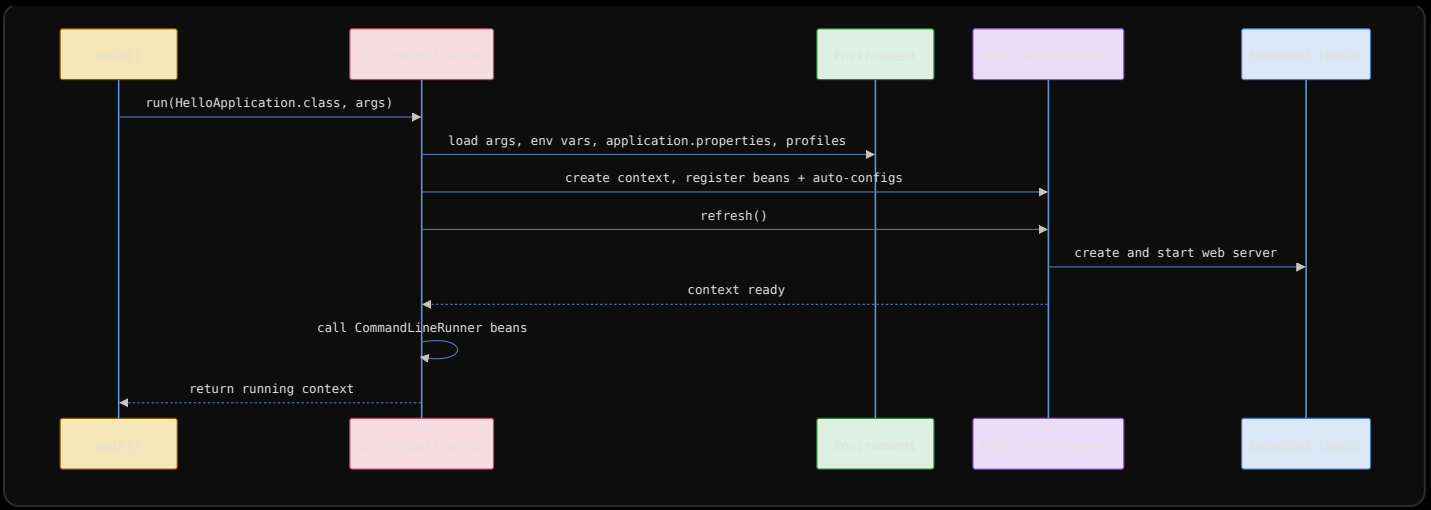
    public static void main(String[] args) {
        SpringApplication.run(HelloApplication.class, args);   // creates and starts the ApplicationContext
    }
}
```

What `@SpringBootApplication` means

Annotation	What it does
`@SpringBootConfiguration`	Marks this class as a configuration source (a specialised `@Configuration`). Tests use it to find your app's configuration.
`@EnableAutoConfiguration`	Loads candidate auto-configuration classes and applies those whose conditions match (Chapter 3).
`@ComponentScan`	Scans **this class's package and all sub-packages** for `@Component`, `@Service`, `@Repository`, `@Controller`, `@RestController`, `@Configuration` classes and registers them as beans.

What `SpringApplication.run` does

1. Decides the application type from the classpath: **servlet** (Spring MVC present), **reactive** (only WebFlux) or **none**.
2. Prepares the `Environment`: command-line args, environment variables, `application.properties`/`.yaml`, active profiles (Chapter 3).
3. Prints the banner.
4. Creates the right `ApplicationContext` (the IoC container; web-server-aware for a servlet app).
5. Registers bean definitions: scanned classes plus auto-configurations whose conditions pass.
6. **Refreshes** the context: creates all singleton beans (one shared instance — the default) and, for a web app, starts embedded Tomcat.
7. Runs `CommandLineRunner`/`ApplicationRunner` beans (run-once-at-startup code).
8. Returns the running context; the JVM keeps running while Tomcat's threads are alive.



Where to put the main class

In the **root package** (`com.example.hello`), with every other class in that package or below (`com.example.hello.orders`, `...web`). Scanning starts from the main class's package, so a class in `com.example.other` is silently **not** a bean.

Common mistakes

- Main class in a sub-package (`com.example.hello.app`) while controllers live in `com.example.hello.web` → endpoints return 404 because the controllers were never scanned.
- Main class in the *default* (unnamed) package — Boot warns, and scanning every jar on the classpath breaks startup.
- Writing logic in `main` after `run(...)` expecting it to run "before the server starts". Use a `CommandLineRunner` or an event listener (Chapter 2).

In the interview

Q: What does `@SpringBootApplication` do? It combines `@SpringBootConfiguration` (this class is configuration), `@EnableAutoConfiguration` (apply Boot's conditional auto-configurations) and `@ComponentScan` (register annotated classes from this package downwards).

- *Follow-up: Scan a package outside the main package?* Move the code under the root package (best), or `@SpringBootApplication(scanBasePackages = "...")` / `@Import` for specific classes.
- *Follow-up: Exclude one auto-configuration?* `@SpringBootApplication(exclude = SomeAutoConfiguration.class)` or the property `spring.autoconfigure.exclude`.

1.9 Running the application

Three ways to run

Way	Command	When
IDE	Run `HelloApplication.main`	Daily development, breakpoints
Maven plugin	`./mvnw spring-boot:run` (or `mvn spring-boot:run`)	Terminal development; reads `src/main/resources` directly
Executable jar	`./mvnw package` then `java -jar target/hello-boot-0.0.1-SNAPSHOT.jar`	Production, containers, CI

Settings on the command line:

```
mvn spring-boot:run -Dspring-boot.run.arguments="--server.port=9090"
java -jar target/hello-boot-0.0.1-SNAPSHOT.jar --server.port=9090 --spring.profiles.active=dev
```

The startup log

Running the packaged jar on JDK 25 (`java -jar target/hello-boot-0.0.1-SNAPSHOT.jar --server.port=18081`):

```
 :: Spring Boot ::                (v4.1.1)      ← after the ASCII-art banner

INFO 91361 --- [hello-boot] [main] c.e.hello.HelloApplication : Starting HelloApplication v0.0.1-SNAPSHOT using
Java 25.0.2 with PID 91361 ...
INFO 91361 --- [hello-boot] [main] c.e.hello.HelloApplication : No active profile set, falling back to 1 default
profile: "default"
INFO 91361 --- [hello-boot] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 18081 (http)
INFO 91361 --- [hello-boot] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache
Tomcat/11.0.24]
INFO 91361 --- [hello-boot] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext:
initialization completed in 376 ms
INFO 91361 --- [hello-boot] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 18081 (http) with
context path '/'
INFO 91361 --- [hello-boot] [main] c.e.hello.HelloApplication : Started HelloApplication in 0.728 seconds (process
running for 0.929)
```

(Timestamps removed, logger names shortened.)

Log part	Meaning
`91361`	Process ID
`[hello-boot]`	`spring.application.name` from `application.properties`
`[main]`	Thread name
`No active profile set...`	Only `application.properties` applies (profiles: Chapter 3)
`Tomcat initialized/started on port ...`	The embedded server is listening
`Started HelloApplication in 0.728 seconds`	The context is fully refreshed and ready

On the first request you also see `Initializing Spring DispatcherServlet 'dispatcherServlet'` — Spring MVC's front controller (one object that receives and routes every request) is created lazily (Chapter 5). Ctrl+C (or `kill <pid>`) shows:

```
INFO ... [ionShutdownHook] o.s.boot.tomcat.GracefulShutdown : Commencing graceful shutdown. Waiting for active
requests to complete
INFO ... [tomcat-shutdown] o.s.boot.tomcat.GracefulShutdown : Graceful shutdown complete
```

Graceful shutdown is on by default: the server stops accepting new requests and lets in-flight ones finish (up to `spring.lifecycle.timeout-per-shutdown-phase`, default 30 s) before the context closes (Chapter 15).

Inside the executable jar

`mvn package` produced a 19.9 MB jar. Layout (`unzip -l`) and manifest (`MANIFEST.MF`, tells Java which class to start):

```
META-INF/MANIFEST.MF
org/springframework/boot/loader/...        ← Boot's launcher classes
BOOT-INF/classes/com/example/hello/*.class ← your code
BOOT-INF/classes/application.properties
BOOT-INF/lib/*.jar                         ← 34 dependency jars, stored unchanged
BOOT-INF/classpath.idx, BOOT-INF/layers.idx
```

```
Main-Class: org.springframework.boot.loader.launch.JarLauncher
Start-Class: com.example.hello.HelloApplication
Spring-Boot-Version: 4.1.1
Spring-Boot-Classes: BOOT-INF/classes/
Spring-Boot-Lib: BOOT-INF/lib/
```

A plain JVM cannot load classes from jars *nested* inside a jar. So `java -jar` starts Boot's **`JarLauncher`** (`Main-Class`), which sets up a class loader able to read `BOOT-INF/lib/*.jar`, then calls your **`Start-Class`**. `layers.idx` groups files into layers (dependencies change rarely, your classes often) so Docker images can cache them (Chapter 15). General jar packaging: Java §18.12.

Common mistakes

- `Web server failed to start. Port 8080 was already in use.` — another process (often a previous run) holds the port. Stop it or set `server.port`; `server.port=0` picks a random free port. Without editing files: `--server.port=9090` or env var `SERVER_PORT`.
- Running `java -jar` on the *original* jar (`*.jar.original`) — no launcher, fails with "no main manifest attribute".
- Expecting DevTools to work with `java -jar`: packaged apps disable it (§1.10).

1.10 DevTools

What it is

Spring Boot DevTools (`spring-boot-devtools`) is a development-only module for a faster edit–run loop:

Feature	What it does
Automatic restart	When compiled classes on the classpath change, the app restarts in about a second.
Development property defaults	Turns off template and static-resource caching, enables the H2 console (web UI for the in-memory H2 database), etc.
Global settings	`~/.config/spring-boot/spring-boot-devtools.properties` applies to all your apps.
Remote DevTools	Opt-in restart of a remote app over HTTP; a security risk — trusted networks only.
LiveReload	Browser auto-refresh; **disabled by default since Boot 4.0, deprecated in 4.1** (no replacement).

How restart works

1. **Two class loaders**: a *base* loader for jars that don't change (Spring, Tomcat, Jackson...) and a *restart* loader for your classes.
2. It watches compiled output folders (e.g. `target/classes`).
3. When something changes there — IDE compile-on-save, IntelliJ *Build Project*, or `mvn compile` — it discards only the restart loader and starts the context again.
4. The warm base loader makes this much faster than a cold start.

Changes under `static/`, `public/`, `templates/` and `META-INF/resources` do **not** trigger a restart. Tune with `spring.devtools.restart.exclude` / `additional-exclude`, or disable with `spring.devtools.restart.enabled=false`.

When it is off

DevTools treats an app launched with `java -jar` (or from a special class loader) as production and disables itself. It is also off in tests: this chapter's test log shows `Restart disabled due to context in which it is running`.

Common mistakes

- Expecting a restart after *editing* a `.java` file without compiling — DevTools watches compiled classes. Enable "build project automatically".
- Shipping DevTools to production by removing `<optional>true</optional>` or configuring the plugin to include it.
- Confusing restart with hot-swapping (replacing code inside the running JVM): restart re-creates the whole context, so in-memory state (sessions, caches) is lost. Tools like JRebel do true class reloading.

1.11 Your first REST endpoint

What we build

`GET /hello?name=Ada` → `{"message":"Hello, Ada!","length":11}`, with the typical layering: a **controller** (HTTP), a **service** (logic) and a **record** (the JSON shape).

```
// Greeting.java — the response body
public record Greeting(String message, int length) {}
```

A **record** is a compact immutable data class (Java 16 — Java §6.8). Jackson 3 serializes its components as JSON fields.

```
// GreetingService.java
@Service                                  // a bean: Spring creates one instance and manages it
public class GreetingService {
    public Greeting greet(String name) {
        String message = "Hello, " + name + "!";
        return new Greeting(message, message.length());
    }
}
```

```
// HelloController.java
@RestController                           // a bean whose method results are written to the HTTP response body
public class HelloController {

    private final GreetingService service;

    HelloController(GreetingService service) { // constructor injection: Spring passes the GreetingService bean
        this.service = service;
    }

    @GetMapping("/hello")                 // handles GET /hello and GET /hello?name=Ada
    Greeting hello(@RequestParam(defaultValue = "World") String name) {
        return service.greet(name);
    }
}
```

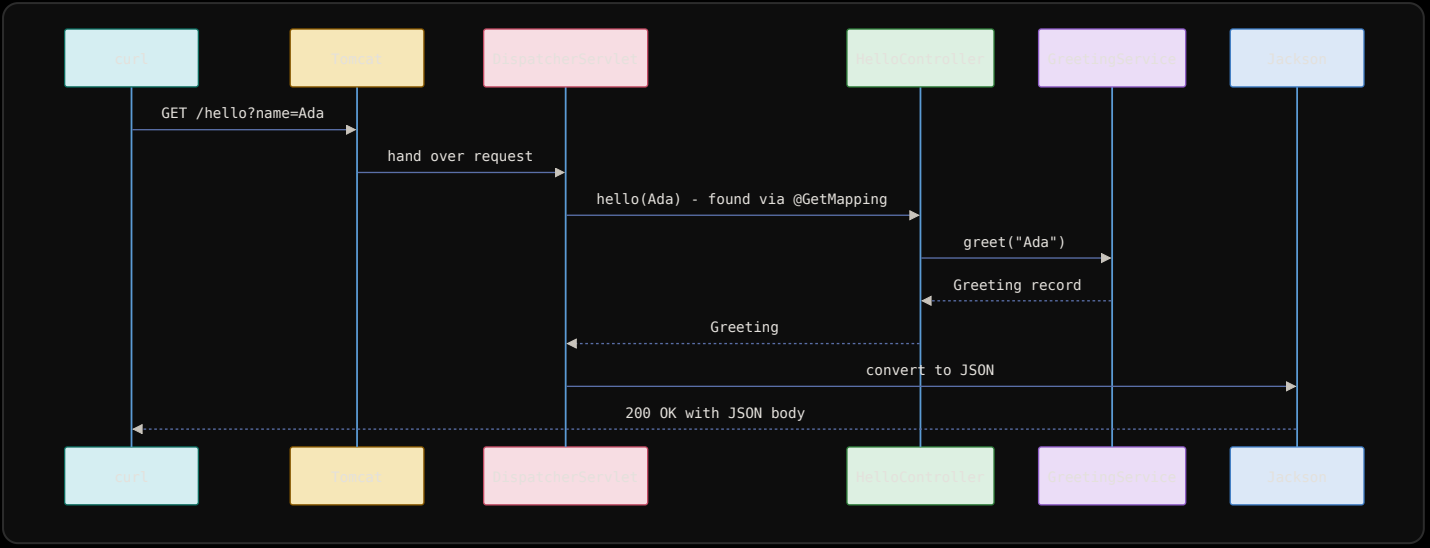
Piece	Role
`@RestController`	`@Controller` + `@ResponseBody`: the return value becomes the response body (JSON), not a view name.
Constructor parameter	Spring passes the singleton `GreetingService`. No `@Autowired` needed with a single constructor (Chapter 2).
`@GetMapping("/hello")`	Maps HTTP GET `/hello` to this method.
`@RequestParam(defaultValue = "World")`	Reads query parameter `name`; if absent, `"World"`.
Return type `Greeting`	Jackson (auto-configured) writes JSON with `Content-Type: application/json`.

Calling it

```
$ curl -s 'localhost:18081/hello?name=Ada'
{"message":"Hello, Ada!","length":11}
$ curl -s localhost:18081/hello
{"message":"Hello, World!","length":13}
$ curl -s -i localhost:18081/hello | head -3
HTTP/1.1 200
Content-Type: application/json
Content-Length: 39
$ curl -s localhost:18081/nope
{"timestamp":"2026-09-28T23:24:50.396Z","status":404,"error":"Not Found","path":"/nope"}
```

The last response is Boot's default error handling (`/error` endpoint). Chapter 5 replaces it with RFC 9457 `ProblemDetail` responses (a standard JSON shape for errors).

What happened during a request



Common mistakes

- `@Controller` instead of `@RestController` for a JSON API → `"Hello"` is treated as a *view name* and fails with a template error.
- Business logic in the controller. Keep controllers thin (parse input, call a service, return output).
- Returning mutable database entities directly — prefer small response records (Chapter 8).

1.12 Testing the first app

Spring Boot tests are ordinary JUnit tests with annotations that start (part of) the container. Two styles cover most needs; Chapter 12 goes deeper. JUnit basics: Java §18.4.

Test 1 — a web *slice* with `MockMvcTester`

```java
// HelloControllerTest.java
@WebMvcTest(HelloController.class)        // only the web layer; no real server
@Import(GreetingService.class)            // the service is not part of the web slice, so add it
class HelloControllerTest {

    @Autowired
    MockMvcTester mvc;                    // AssertJ-flavoured MockMvc (Framework 6.2+)

    @Test
    void greetsWorldByDefault() {
        var result = mvc.get().uri("/hello").exchange();
        System.out.println("GET /hello          -> " + result.getResponse().getStatus() + " " +
contentOf(result));
        assertThat(result).hasStatusOk()
                .bodyJson().extractingPath("$.message").isEqualTo("Hello, World!");
    }

    @Test
    void greetsByName() {
        var result = mvc.get().uri("/hello?name=Ada").exchange();
        System.out.println("GET /hello?name=Ada -> " + result.getResponse().getStatus() + " " +
contentOf(result));
        assertThat(result).hasStatusOk()
                .bodyJson().extractingPath("$.length").isEqualTo(11);
    }
    // contentOf(...) reads the response body as a String
}
```

- `@WebMvcTest` starts a *slice*: MVC infrastructure, controllers, JSON and filters only — no services, repositories or server. Hence `@Import(GreetingService.class)` (the common alternative is mocking it with `@MockitoBean`, Chapter 12).
- `MockMvcTester` sends requests straight to the `DispatcherServlet` in memory, with AssertJ assertions like `hasStatusOk()` and `bodyJson().extractingPath(...)` (JSONPath; `$` is the root).

Test 2 — the whole app on a real port with `RestTestClient`

```java
// HelloApplicationTests.java
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.RANDOM_PORT)
@AutoConfigureRestTestClient
class HelloApplicationTests {

    @LocalServerPort int port;
    @Autowired RestTestClient client;      // Framework 7 test client, pre-bound to the random port

    @Test
    void realHttpCall() {
        String body = client.get().uri("/hello?name=Spring")
                .exchange()
                .expectStatus().isOk()
                .expectHeader().contentType("application/json")
                .expectBody(String.class).returnResult().getResponseBody();
        System.out.println("Tomcat on random port " + (port > 0 ? "OK" : "?") + ", body = " + body);
        assertThat(body).isEqualTo("{\"message\":\"Hello, Spring!\",\"length\":14}");
    }
}
```

- `@SpringBootTest` starts the full context like `main`; with `RANDOM_PORT` it also starts Tomcat on a free port, so the test makes real HTTP calls.
- `RestTestClient` (new in Framework 7) is a fluent test client for blocking apps; `@AutoConfigureRestTestClient` (module `spring-boot-resttestclient`, included by the webmvc test starter) binds it to the running server.
- Surefire (the unit-test plugin) runs only classes ending in `Test`, `Tests` or `TestCase` by default — a class like `HelloApplicationIT` never runs with `mvn verify` here; `*IT` classes need the Failsafe plugin, which this project does not configure.

Output

`mvn -q -B -ntp verify` (banners and log lines removed):

```
Tomcat on random port OK, body = {"message":"Hello, Spring!","length":14}
GET /hello?name=Ada -> 200 {"message":"Hello, Ada!","length":11}
GET /hello          -> 200 {"message":"Hello, World!","length":13}
```

(JUnit does not guarantee method order, so the two `GET` lines can swap.) Exit code 0: all three tests passed.

Trade-offs

	`@WebMvcTest` + `MockMvcTester`	`@SpringBootTest(RANDOM_PORT)` + `RestTestClient`
Starts	Web layer only, no server	Whole app + real Tomcat
Speed	Fast	Slower
Catches	Mapping, JSON, validation, controller logic	Everything incl. filters, server config, real HTTP
Use for	Most controller tests	A few end-to-end "smoke" tests

1.13 In the interview

Q: Walk me through what happens when you run a Spring Boot jar. `java -jar` starts `JarLauncher` from the manifest; it builds a class loader for `BOOT-INF/lib` jars and calls the `Start-Class` main method. `SpringApplication.run` detects a servlet app, prepares the `Environment` (args, env vars, config files, profiles), creates the `ApplicationContext`, registers scanned beans and matching auto-configurations, refreshes the context — creating singletons and starting embedded Tomcat — then runs `CommandLineRunner`s.

- *Follow-up: Why an embedded server?* One self-contained artifact per service: same jar in dev, CI and production, easy to containerise, no server-version drift.
- *Follow-up: Can you still build a WAR?* Yes: packaging `war`, extend `SpringBootServletInitializer`, mark the server starter `provided`.

Q: What is a starter? A code-less dependency bringing a curated, version-aligned library set plus the Boot module holding its auto-configuration; in Boot 4, one per technology with a `-test` companion (§1.7).

Q: What changed in Spring Boot 4 / Framework 7? Java 17 baseline (25 recommended), Jakarta EE 11 (Servlet 6.1, Tomcat 11), modularized Boot jars and renamed starters, Jackson 3 (`tools.jackson`), JUnit 6, JSpecify null-safety, API versioning in MVC/WebFlux, HTTP service client auto-configuration, `RestTestClient`, core retry/concurrency-limit annotations, Undertow removed. Details in Chapter 17.

Q: `@WebMvcTest` vs `@SpringBootTest`? Fast MVC-only slice vs the whole app (optionally on a real server) — see the §1.12 trade-offs table.

Chapter TL;DR

- **Spring Framework** = IoC container that creates and wires your objects (beans) + modules for web, data, transactions, messaging.
- **Spring Boot** = Spring + starters + managed versions (BOM) + auto-configuration + embedded server + externalized config + Actuator.
- **Current generation**: Framework 7 and Boot 4 (Nov 2025); this book uses **Boot 4.1.1 → Framework 7.0.9**, Jakarta EE 11, Tomcat 11, Jackson 3, JUnit 6, on **Java 25** (Java 17 minimum).
- Generate projects at **start.spring.io**; the `pom.xml` inherits `spring-boot-starter-parent`, has version-less starters and the `spring-boot-maven-plugin`.

- **Boot 4 starters are modular**: `spring-boot-starter-webmvc` (was `-web`), a `-test` starter per technology.
- `@SpringBootApplication` = `@SpringBootConfiguration` + `@EnableAutoConfiguration` + `@ComponentScan` (from the main class's package down).
- Run with the IDE, `mvn spring-boot:run`, or `java -jar` on the fat jar (`JarLauncher` → your `Start-Class`). Graceful shutdown is on by default.
- **DevTools** restarts fast on recompilation and disables itself for packaged jars and tests.
- A REST endpoint is a `@RestController` method with `@GetMapping`; return a record and Jackson writes JSON.
- Test controllers with `@WebMvcTest` + `MockMvcTester`; smoke-test the whole app with `@SpringBootTest(RANDOM_PORT)` + `RestTestClient`.

Next: Chapter 2 — IoC and Dependency Injection opens up the container you just used.